

Dosimetry 2

Oral Exam Master Packet

VMAT, IMRT, 3D planning, dose limits, re-irradiation, CSI, and classic field setups

Purpose

This packet is written to be the kind of document you can *study from directly* and also *speak from in an oral exam*. It is intentionally organized around questions examiners tend to ask:

- When do you choose 3DCRT, static IMRT, or VMAT?
- What makes VMAT commissioning and QA different?
- What are the important dose limits by body site?
- How do serial and parallel organs behave differently?
- How do dose limits change in re-irradiation or palliative cases?
- What is EUD, and what does the a parameter mean?
- How do you set up CSI, mantle fields, and field junctions?
- What dose grid and calculation-point logic should you use?

How to use this packet

Read the blue and green boxes first. Memorize the site-specific dose anchors, then practice the rapid-fire oral exam answers out loud until the wording feels natural.

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High-yield oral exam review for clinical dosimetry, treatment planning, and delivery decision-making

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1 How to Answer This Oral Exam Well

High-yield point

The strongest oral answers are usually:

- principle first, number second;
- site-specific, not generic;
- risk-aware, not cookbook;
- honest about uncertainty;
- clear about whether the case is definitive, re-irradiation, or palliative.

Oral-exam answer

How should you approach a planning technique question?

A strong answer is:

I first define the target priority, then classify critical OARs as serial or parallel, then choose the simplest planning technique likely to meet those priorities. I evaluate coverage volumetrically with ICRU-style metrics and only add IMRT or VMAT complexity when it improves the therapeutic ratio.

Clinical workflow

General planning workflow

1. Confirm prescription, intent, and fractionation.
2. Review immobilization, simulation, and image guidance needs.
3. Review target volumes and OARs.
4. Classify critical OARs as serial, parallel, or mixed.
5. Choose technique: 3DCRT, static IMRT, or VMAT.
6. Choose beam arrangement / arc arrangement.
7. Select clinical dose grid and algorithm.
8. Evaluate target coverage, conformity, homogeneity, and OAR constraints.
9. Perform secondary dose/MU verification and PSQA if required.
10. Confirm deliverability, clearance, imaging, and documentation.

Common trap

The oral-exam trap: “The best plan is the most conformal plan.”

That is wrong. The best plan is the plan that safely meets the clinical goal while remaining robust, deliverable, and appropriate for the patient’s intent of treatment.

2 3DCRT, Static IMRT, and VMAT

2.1 Forward versus Inverse Planning

Definition

Forward planning

Forward planning means the planner manually chooses field angles, beam weights, wedges, blocks, MLC shapes, and segments, then iteratively improves the dose. This is the classic 3DCRT and field-in-field mindset.

Definition

Inverse planning

Inverse planning means the planner defines target objectives and OAR constraints, and the optimizer solves for fluence, aperture weights, or machine parameters. IMRT and VMAT are typically inverse planned.

Oral-exam answer

When is forward planning still the right answer?

Forward planning is still appropriate when:

- geometry is simple,
- OAR overlap is limited,
- modulation provides little clinical benefit,
- the case is palliative or simple,
- low-dose bath should be minimized,
- robustness and simplicity matter more than tight conformity.

2.2 Technique Comparison

Table 1: High-yield comparison of 3DCRT, static IMRT, and VMAT.

Technique	Planning style	Advantages	Limitations
3DCRT	Forward planned	Simple, robust, fast, easier to verify, often less low-dose bath	Less conformal for complex or concave targets
Static IMRT	Inverse planned, fixed beams	Directional modulation, strong OAR sparing, useful when beam direction matters	Longer delivery, more MU, more segments
VMAT	Inverse planned, rotating arcs	Efficient, conformal, shorter treatment time, strong for SIB workflows	Broader low-dose bath possible; dynamic delivery complexity

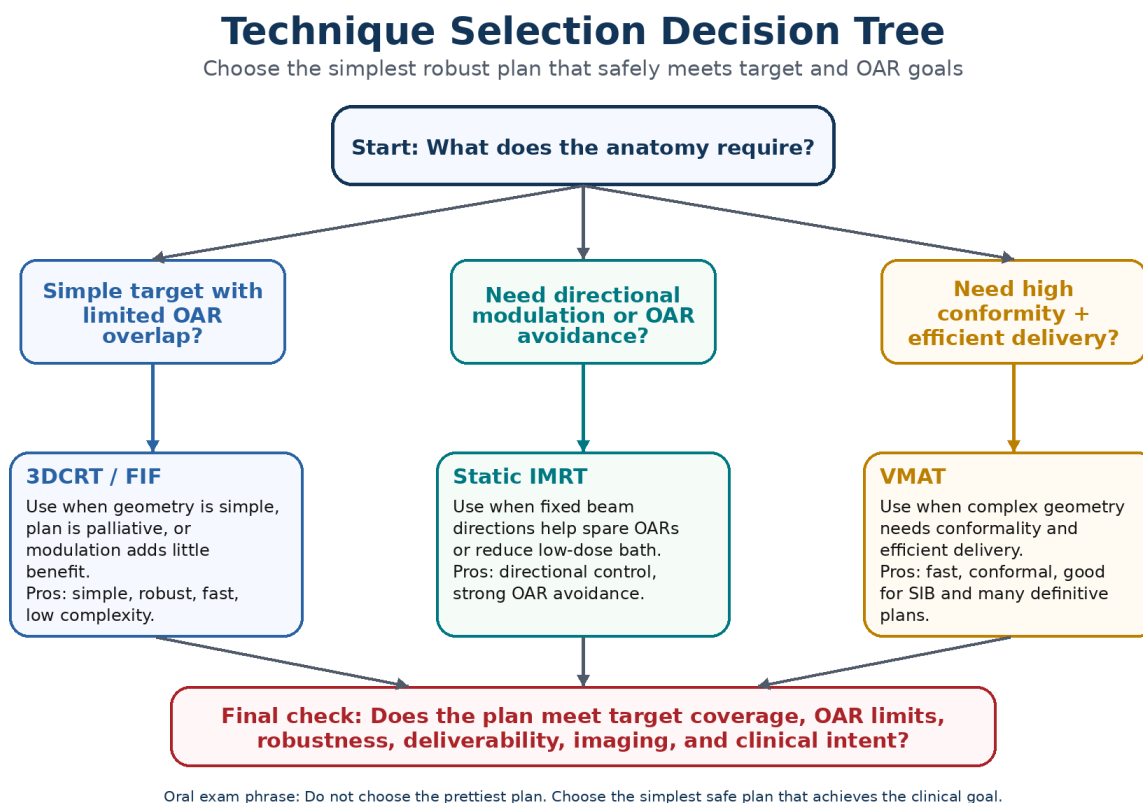


Figure 1: Technique selection decision tree for choosing between 3DCRT, static IMRT, and VMAT.

Oral-exam answer

How do you choose between 3DCRT, static IMRT, and VMAT?

If the target is simple and OAR overlap is limited, I prefer 3DCRT because it is simple and robust. If I need directional modulation or want to avoid spreading low dose through large volumes, I consider static IMRT. If I need high conformity, efficient delivery, or SIB planning, I consider VMAT.

Oral-exam answer

When might static IMRT beat VMAT?

Static IMRT can be better when:

- directional control matters more than speed,
- low-dose bath from arcs is undesirable,
- partial beam avoidance is needed,
- fixed-field delivery is easier to verify or more robust.

Oral-exam answer

When might 3DCRT beat both IMRT and VMAT?

3DCRT may be best for:

- simple palliative plans,

- uncomplicated target geometry,
- low-resource or urgent situations,
- cases where modulation adds low-dose bath without meaningful clinical benefit.

3 VMAT Criteria, Commissioning, and QA

Definition

VMAT

VMAT is rotational IMRT. During delivery, the gantry rotates while the machine simultaneously modulates MLC positions, dose rate, and gantry speed.

High-yield point

One-line exam answer

VMAT is not just IMRT delivered faster. It couples MLC motion, gantry motion, and dose-rate modulation.

3.1 VMAT Plan Quality Criteria

Oral-exam answer

What criteria do you evaluate for a VMAT plan?

For VMAT, I evaluate:

- target coverage,
- conformity,
- homogeneity,
- dose gradient,
- OAR constraints,
- low-dose bath,
- modulation complexity,
- deliverability,
- patient-specific QA,
- secondary dose verification.

Key formula

Common plan quality metrics

$$CI = \frac{\text{Prescription isodose volume}}{\text{Target volume}}$$

$$HI \approx \frac{D_{2\%} - D_{98\%}}{D_{50\%}}$$

Lower *CI* deviation and lower *HI* generally indicate better conformity and homogeneity, but clinical interpretation is site-dependent.

3.2 VMAT Commissioning / Validation

Clinical workflow

VMAT-specific commissioning checklist

- Validate MLC position accuracy during gantry rotation.
- Validate variable dose-rate delivery.
- Validate gantry-speed modulation.
- Validate leaf-speed limits.
- Validate small-MU control points.
- Validate control-point interpolation.
- Validate arc start, stop, interruption, and resume behavior.
- Validate plan transfer to R&V and delivery system.
- Validate with TG-119-style or representative clinical cases.
- Perform end-to-end testing with measurement.

Oral-exam answer

Why is VMAT harder to QA than static IMRT?

VMAT varies MLC position, gantry angle, gantry speed, and dose rate at the same time. A static IMRT field tests modulation at fixed gantry positions, but VMAT tests whether the machine can coordinate modulation during motion.

3.3 Patient-Specific QA

High-yield point

Common TG-218-style VMAT QA criteria are often summarized as:

$$3\%/2 \text{ mm}, \quad 10\% \text{ threshold}, \quad \geq 95\% \text{ pass tolerance}$$

with an action level around 90%, depending on local policy.

Common trap

Classic pitfall

Do not say: “*The VMAT plan is good because the gamma was 99%.*”

Gamma is a QA screen. Plan quality still depends on target coverage, OAR DVH, deliverability, image guidance, robustness, and clinical intent.

Oral-exam answer

A VMAT plan passed gamma. Are you done?

No. I still need physician-approved target coverage and OAR DVHs, independent dose/MU verification, correct prescription interpretation, machine QA status, image guidance feasibility, and deliverability review.

4 Isocenter, Calculation Point, and Dose Grid

4.1 Isocenter versus Calculation Point

Definition**Isocenter**

The isocenter is the geometric reference point where gantry, collimator, and couch rotations intersect. It is mainly a delivery and setup reference.

Definition**Calculation point**

The calculation point is a dosimetric sampling point used for point-dose checks, MU checks, or normalization in some workflows.

Oral-exam answer

Is the isocenter always the prescription point?

No. In modern IMRT/VMAT, the prescription is usually volume-based, not point-based. The isocenter is often chosen for setup and delivery geometry. If I use a calculation point, I choose a high-dose, low-gradient, clinically representative region.

4.2 Dose Grid Requirements

Oral-exam answer

What dose grid should you use?

There is no universal grid size for all sites. The dose grid should match the clinical use case and must be validated. Routine IMRT/VMAT commonly uses about 2–3 mm. SRS/SBRT, small targets, or steep gradients may require 1–2 mm.

Common trap**Common trap**

Do not answer the dose-grid question with one magic number only. A better answer is:

Use the clinically validated grid. Routine IMRT/VMAT may be 2–3 mm, but steep gradients or small targets require finer resolution.

5 Serial and Parallel Organs

Definition

Serial organ

A serial organ can lose function if a small critical subvolume receives too much dose. Serial organs are hot-spot sensitive.

Definition

Parallel organ

A parallel organ has functional subunits distributed throughout the volume. Parallel organs are usually mean-dose or volume-dose sensitive.

Organ type	Examples	Planning metric
Serial	Cord, brainstem, optic chiasm, optic nerves	Maximum dose or near-maximum dose
Parallel	Parotids, lung, liver, kidney	Mean dose, Vx, spared volume
Mixed / endpoint dependent	Bowel, esophagus, heart	Depends on endpoint and volume involved

Table 2: Serial versus parallel organ behavior.

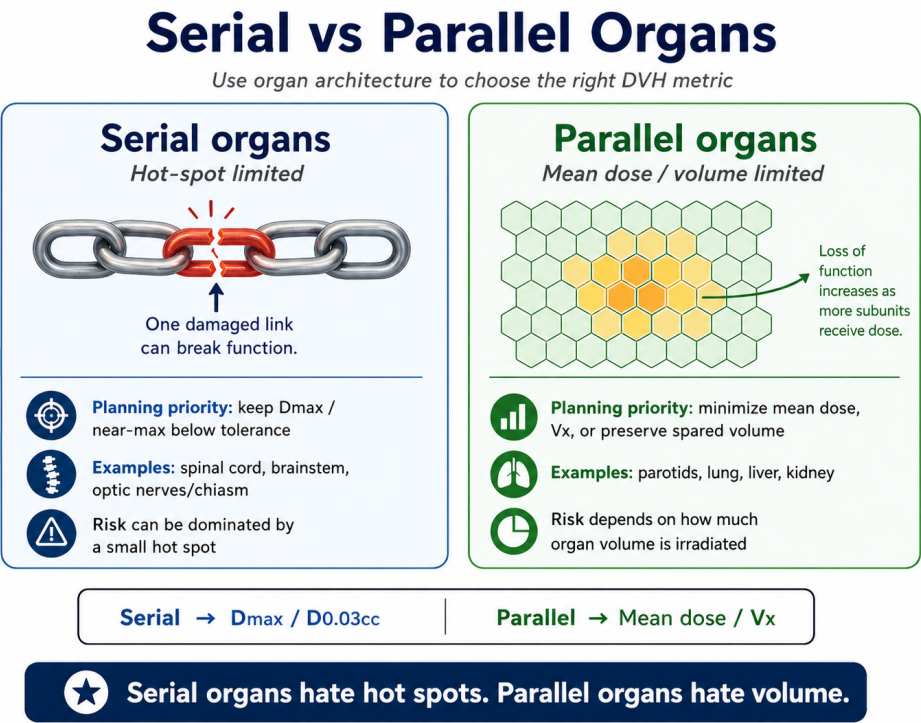


Figure 2: Conceptual comparison of serial and parallel organ behavior. Serial organs are primarily maximum-dose or near-maximum-dose limited, while parallel organs are more sensitive to mean dose, irradiated volume, or spared functional volume.

Oral-exam answer

Why does serial versus parallel matter in planning?

It tells me which DVH metric to prioritize. For a serial organ, I worry about focal hot spots. For a parallel organ, I care more about mean dose and irradiated volume. That changes both optimization priorities and technique selection.

High-yield point

Memory phrase

Serial organs hate hot spots. Parallel organs hate volume.

6 EUD and the Alpha Parameter

Definition

Equivalent Uniform Dose

EUD is the uniform dose that would produce the same biological effect as a nonuniform dose distribution.

Key formula

Generalized EUD

$$\text{gEUD} = \left(\sum_i v_i D_i^a \right)^{1/a}$$

where v_i is fractional volume receiving dose D_i , and a controls volume sensitivity.

Table 3: Interpretation of the EUD a parameter.

a behavior	Interpretation	Clinical meaning
$a = 1$	EUD behaves like mean dose	Parallel-like behavior
Large positive a	EUD approaches maximum dose	Serial-organ / hot-spot sensitivity
Negative a	EUD emphasizes cold spots	Target underdose sensitivity

Oral-exam answer**What does the EUD a parameter mean?****Example of how to use gEUD**

Suppose an organ has the following dose distribution:

30% of the volume gets 10 Gy

50% of the volume gets 20 Gy

20% of the volume gets 30 Gy

Then:

$$\text{gEUD} = (0.30(10)^a + 0.50(20)^a + 0.20(30)^a)^{1/a}$$

If $a = 1$, then gEUD becomes the mean dose:

$$\text{gEUD} = 0.30(10) + 0.50(20) + 0.20(30)$$

$$\text{gEUD} = 3 + 10 + 6 = 19 \text{ Gy}$$

So when $a = 1$, gEUD is just the average dose.**EUD Alpha Parameter***How the volume-effect parameter a changes what the DVH metric cares about.*

$$\text{gEUD} = (\sum v_i D_i^a)^{1/a} \quad \bigg| \quad \sum v_i = 1$$

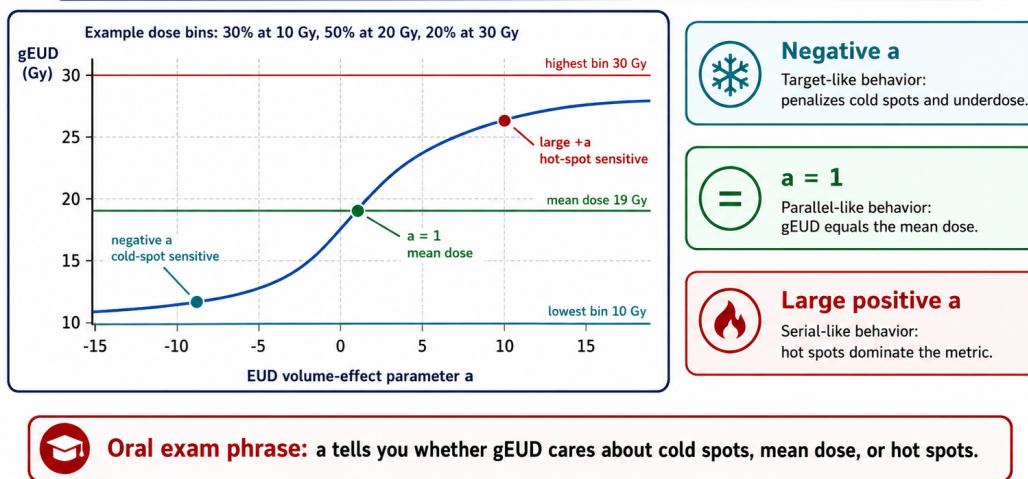
 v_i = fractional volume in dose bin i D_i = dose to that bin a = volume-effect parameter

Figure 3: Conceptual effect of the generalized EUD volume-effect parameter a . Negative a values emphasize cold spots, $a = 1$ gives the mean dose, and large positive a values increasingly emphasize hot spots.

Structure / endpoint	Typical a	Behavior	Oral-exam interpretation
Target coverage / tumor control	-10 to -20	Cold-spot sensitive	Negative a emphasizes underdose. A small cold spot can reduce tumor control.
Spinal cord	~ 20	Strongly serial	Maximum or near-maximum dose dominates risk.
Optic nerve / chiasm	~ 10 to 20	Serial	Small high-dose regions can cause major toxicity.
Brainstem	~ 10 to 12	Serial	Primarily constrained by maximum or near-maximum dose.
Esophagus	~ 10 to 20	Serial / mixed	Endpoint dependent. Stricture behaves more serially than broad acute irritation.
Rectum	~ 8	Mixed, high-dose sensitive	High-dose volume matters; think $V70$, $V75$, and the high-dose DVH tail.
Bladder	~ 2 to 3	Mixed / volume effect	Volume effect is important, but high-dose regions still matter clinically.
Femoral head	~ 4	Mixed	Avoid focal high dose, but it is not as purely serial as cord or optic structures.
Parotid gland	~ 1	Parallel	Mean dose is the key metric for xerostomia risk.
Lung	~ 1	Parallel	Mean lung dose and $V20$ are clinically important.
Liver	~ 1	Parallel	Mean liver dose and spared functional liver volume dominate.
Kidney	~ 1	Parallel	Mean dose and spared kidney volume are more important than a small hot spot.

Table 4: Typical generalized EUD a -parameter values by structure or endpoint. These values are approximate modeling examples and should not be treated as universal clinical constraints.

Common trap**Important EUD caveat**

The a -parameter is a **model parameter**, not a hard clinical dose constraint. Different publications, endpoints, and fitting methods may give different values. The important oral-exam concept is the trend:

$$a < 0 \Rightarrow \text{cold-spot sensitive target behavior}$$

$$a \approx 1 \Rightarrow \text{mean-dose / parallel-organ behavior}$$

$$a \gg 1 \Rightarrow \text{hot-spot / serial-organ behavior}$$

The parameter a tells me what part of the DVH the metric cares about. Large positive a emphasizes hot spots and behaves more like a serial organ. $a \approx 1$ behaves like mean dose and is more parallel-like. Negative a emphasizes cold spots, which is useful for target underdose.

7 Dose Limits by Body Site

Common trap**Important caution**

The following are representative oral-exam anchors, mostly for conventional fractionation. They are not universal laws. Actual clinical constraints depend on prescription, fractionation, protocol, prior radiation, concurrent therapy, physician preference, and institutional policy.

7.1 Head and Neck

Oral-exam answer**What dose limits matter in head and neck?**

Important OARs include:

- spinal cord,
- brainstem,
- parotids,
- oral cavity,
- pharyngeal constrictors,
- larynx,
- mandible,
- optic structures for skull base cases.

High-yield point

Representative anchors:

- Spinal cord: keep around 45–50 Gy in conventional fractionation.
- Brainstem: around 54 Gy conventional anchor.
- Parotid: one gland mean < 20 Gy or both means < 25 Gy when feasible.
- Definitive gross disease: commonly around 70 Gy.
- Elective nodal dose: commonly around 54–57 Gy depending on SIB scheme.

7.2 Thorax

Oral-exam answer

What dose limits matter in thorax?

Important OARs include:

- total lung minus GTV,
- spinal cord,
- heart,
- esophagus,
- brachial plexus for apical disease.

High-yield point

Representative anchors:

- Lung: $V20 \leq 30\text{--}35\%$
- Mean lung dose: < 20–23 Gy
- Cord: often ≤ 50 Gy
- Stage III NSCLC anchor prescription: 60 Gy in 30 fractions

7.3 Prostate / Pelvis

Oral-exam answer

What dose limits matter in prostate and pelvis?

Important OARs include:

- rectum,
- bladder,
- bowel bag / small bowel,
- femoral heads,
- penile bulb when clinically relevant.

High-yield point

Representative anchors:

- Prostate definitive conventional: 76–80 Gy
- Moderate hypofractionation example: 60 Gy in 20 fractions
- Rectum: high-dose tail matters, such as V_{70} , V_{75}
- Bladder: high-dose volume matters, such as V_{65} , V_{70} , V_{75}
- Bowel bag: $V_{45} < 195$ cc is a common oral anchor

7.4 Abdomen

Oral-exam answer

What dose limits matter in abdomen?

Abdomen is highly site-specific. Important OARs include:

- bowel / bowel bag,
- kidneys,
- liver,
- stomach,
- cord,
- duodenum for pancreatic/upper abdominal cases.

High-yield point

Representative anchors:

- Small bowel loops: $V_{15} < 120$ cc
- Peritoneal cavity / bowel bag: $V_{45} < 195$ cc
- Kidneys: mean dose matters; try to spare functional kidney volume
- Liver: mean dose and spared liver volume matter

7.5 Brain

Oral-exam answer

What dose limits matter in brain?

Important OARs include:

- optic nerves,
- optic chiasm,
- brainstem,

- cochlea,
- normal brain,
- hippocampi for select whole-brain cases.

High-yield point

Representative anchors:

- Glioblastoma: 60 Gy in 30 fractions
- Whole brain: 30 Gy in 10 fractions
- Brainstem: around 54 Gy conventional anchor
- Optic nerve/chiasm: often around 54–55 Gy conventional anchor
- SRS: normal brain V12, conformity, and gradient become key

Site	Key OARs	Oral-exam anchors
Head & Neck	Cord, brainstem, parotids, oral cavity, mandible	Cord 45–50 Gy, brainstem ~ 54 Gy, parotid mean < 20 –25 Gy, gross disease ~ 70 Gy.
Thorax	Lung, cord, heart, esophagus	Lung $V20 \leq 30$ –35%, mean lung dose < 20 –23 Gy, cord ≤ 50 Gy, NSCLC 60 Gy/30.
Prostate / Pelvis	Rectum, bladder, bowel, femoral heads	Prostate 76–80 Gy conventional or 60 Gy/20; rectum/bladder high-dose volumes; bowel bag $V45 < 195$ cc.
Abdomen	Bowel, kidneys, liver, stomach, duodenum	Abdomen is site-specific; bowel bag $V45 < 195$ cc, loops $V15 < 120$ cc, kidney/liver mean dose and spared volume matter.
Brain	Brainstem, optic nerves/chiasm, cochlea, normal brain	GBM 60 Gy/30, WBRT 30 Gy/10, brainstem ~ 54 Gy, optic structures ~ 54 –55 Gy.

Table 5: High-yield dose-limit anchors by site for oral exam review.

8 Re-Irradiation and Palliative Cases

8.1 Re-Irradiation

Oral-exam answer

How do you handle dose limits for re-irradiation?

There is no universal safe cumulative dose limit. I would review the prior dose distribution, evaluate overlap with the new plan, convert to BED/EQD2 if useful, consider time interval and tissue recovery, and prioritize serial OARs such as cord, brainstem, and optic structures.

Key formula

BED and EQD2

$$BED = nd \left(1 + \frac{d}{\alpha/\beta} \right)$$

$$EQD2 = D \frac{d + \alpha/\beta}{2 + \alpha/\beta}$$

Common trap

Classic pitfall

Do not quote a single universal re-irradiation number. Re-irradiation tolerance depends on prior dose, fractionation, overlap, endpoint, organ architecture, time interval, and clinical intent.

8.2 Palliative Cases

Oral-exam answer

Do dose limits matter in palliative cases?

Yes. Palliation changes the balance between simplicity, speed, symptom relief, and long-term toxicity risk, but it does not eliminate OAR constraints. Serial organs such as cord and brainstem still matter.

High-yield point

Common palliative prescriptions:

- Bone metastasis: 8 Gy $\times$ 1
- Bone metastasis: 20 Gy in 5 fractions
- Bone metastasis: 30 Gy in 10 fractions
- Whole brain: 30 Gy in 10 fractions

9 CSI and Mantle Fields

9.1 Craniospinal Irradiation

Definition

CSI

Craniospinal irradiation treats the entire craniospinal axis, typically using opposed lateral cranial fields matched to one or more posterior spine fields.

Clinical workflow

Classic CSI setup

- Opposed lateral whole-brain fields.
- One or two posterior spine fields.
- Match cranial and spinal fields carefully.
- Use couch / collimator / gantry rotations to manage divergence.
- Feather junctions periodically.

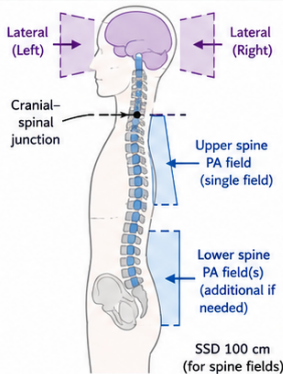
Craniospinal Irradiation (CSI): Setup, Junction Matching, and Feathering

Opposed lateral brain fields matched to posterior spine fields, with junction shifts used to spread setup error.

1 Classic CSI setup

Simulation:

- Supine or prone; shoulders down; neck extended; spine straight
- Thermoplastic mask over head/neck/shoulders
- Vac bag for body; knee sponge can reduce lordosis

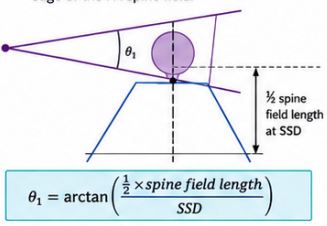


- Typically need 3 fields.
- Plan upper spine field first.
- Whole brain field = 2 opposed laterals.
- Lower spine field(s) = additional PA field(s).

2 Matching brain and spine fields

A Collimator rotation on lateral brain fields

Rotate collimator on both opposed laterals to match the upward divergence of the superior edge of the PA spine field.

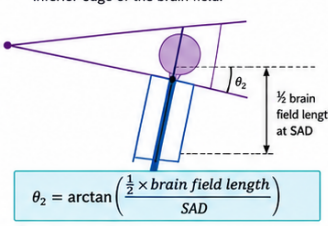


$\theta_1 = \arctan\left(\frac{\frac{1}{2} \times \text{spine field length}}{\text{SSD}}\right)$

Whole brain field: Rotate collimator on opposed laterals About 12° (geometry dependent)

B Couch kick on the spine field

Kick the couch TOWARD the head of gantry to match the downward divergence of the inferior edge of the brain field.



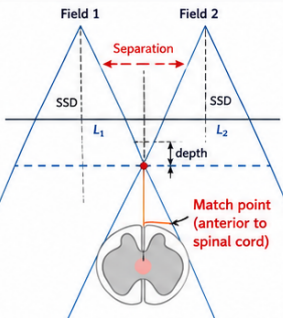
$\theta_2 = \arctan\left(\frac{\frac{1}{2} \times \text{brain field length}}{\text{SAD}}\right)$

Spine field: Couch kick toward head of gantry About 8° (geometry dependent)

Notes: SSD = source-skin distance (spine). SAD = source-axis distance (brain). Lengths measured at skin.

3 Matching multiple spine fields

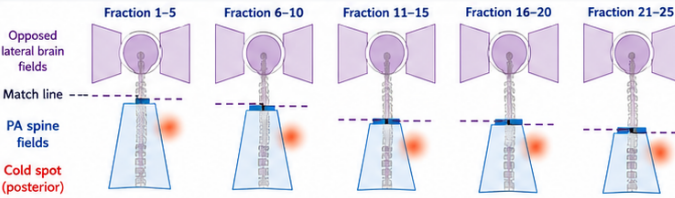
- Match spine field ANTERIOR to spinal cord to avoid hot spot in cord.
- Therapists can place a skin marker at the match point to check the gap.
- Calculate skin separation using similar triangles.



$$\text{Separation} = \frac{1}{2} \times L \times \left(\frac{\text{depth}}{\text{SSD}}\right)$$

SSD = source-skin distance
L = spine field length at skin
depth = depth of match point anterior to cord

4 Feathering the junction



- A fixed junction repeats the same hot/cold spot at the same spinal level every day.
- Move the match line periodically to spread setup/output mismatch over a longer spinal region.
- Typical total shift range is about ± 1 to ± 2 cm.
- Move the junction every 5 fractions or weekly depending on protocol.
- Therapists should verify marker position at each new match level.

Oral exam pearl: Feathering intentionally moves the match line so setup or output mismatch is not repeated at the same spinal level every day.

Figure 4: Craniospinal irradiation setup, brain-spine field matching, multiple spine-field matching, and junction feathering. Opposed lateral brain fields are matched to posterior spine fields using collimator rotation, couch kick, field separation, and periodic junction shifts to reduce repeated hot or cold spots at the same spinal level.

Oral-exam answer

What is the purpose of feathering in CSI?

Feathering shifts the junction position during treatment so that small setup errors do not create the same hot or cold spot every day. Instead of a fixed overdose or underdose band, the error is spread over a larger region.

High-yield point

Representative CSI prescriptions:

- Average-risk medulloblastoma: 23.4 Gy CSI plus boost to approximately 54–55.8 Gy
- High-risk medulloblastoma: 36 Gy CSI plus boost per protocol

9.2 Mantle Fields**Definition****Mantle field**

Mantle field irradiation is a historical extended-field technique used for supradiaphragmatic Hodgkin lymphoma nodal irradiation.

Oral-exam answer**What does a mantle field treat?**

A classic mantle field treats cervical, supraclavicular, infraclavicular, mediastinal, hilar, and axillary lymphatic regions depending on the exact mantle variant.

Clinical workflow**Classic mantle setup**

- Large AP/PA supradiaphragmatic fields.
- Custom lung blocks.
- Possible larynx block.
- Possible humeral head shielding.
- Arm position chosen to balance axillary coverage and normal tissue sparing.
- Large irregular field dosimetry and heterogeneity are major concerns.

Common trap**Oral-exam nuance**

Mantle fields are largely historical, but they test your ability to think about large-field geometry, shielding, heterogeneity, field borders, and normal tissue compromise.

10 Troubleshooting Planning Problems**Common trap****Parotids will not meet goal**

First check whether one-gland sparing is clinically realistic. Then review overlap with nodal targets, arc path, beam direction, objective strength, and whether target coverage is being over-prioritized.

Common trap**Lung V20 is acceptable, but V5 is too broad**

Consider fewer arcs, partial arcs, static IMRT, beam-angle restriction, or less modulation. If low-dose bath is the problem, regain directionality.

Common trap**Cord or brainstem hot spot**

For a serial organ, a small hot spot can be clinically decisive. Re-prioritize the serial OAR, reassess normalization, evaluate contour/margin geometry, and consider changing technique.

Common trap**VMAT QA fails despite acceptable DVH**

Determine whether the failure is a measurement issue, setup issue, modeling issue, delivery issue, or plan complexity issue. Do not blindly replan before identifying the failure mode.

11 Rapid-Fire Oral Exam Answers

11.1 The shortest answers you should be able to give cleanly

Oral-exam answer**3DCRT versus IMRT versus VMAT**

3DCRT is simple, robust, and forward planned. IMRT uses fixed-field inverse modulation. VMAT is rotational IMRT with simultaneous MLC motion, gantry motion, and dose-rate modulation.

Oral-exam answer**When do you use VMAT?**

I use VMAT when I need high conformity and efficient delivery, especially for complex targets, SIB plans, prostate, head and neck, and many modern definitive cases.

Oral-exam answer**When do you avoid VMAT?**

I avoid or limit VMAT when the low-dose bath is unacceptable, when directional avoidance is more important than delivery speed, or when motion/robustness makes sharp modulation unsafe.

Oral-exam answer**VMAT QA criteria**

A common TG-218-style answer is global gamma 3%/2 mm, 10% threshold, with about 95% passing as a tolerance and 90% as an action level, depending on local policy.

Oral-exam answer**Serial versus parallel organs**

Serial organs are hot-spot sensitive, so I watch max dose. Parallel organs are volume/mean-dose sensitive, so I watch mean dose, Vx, and spared volume.

Oral-exam answer**Cord constraint**

For conventional fractionation, a practical spinal cord planning anchor is around 45–50 Gy, depending on contouring, fractionation, and institutional policy.

Oral-exam answer**Parotid constraint**

Parotids are parallel organs. A common anchor is one parotid mean < 20 Gy or both parotids mean < 25 Gy when feasible.

Oral-exam answer**Thorax constraints**

For lung, I think about total lung minus GTV, V_{20} , and mean lung dose. Common anchors are $V_{20} \leq 30$ –35% and mean lung dose < 20 –23 Gy.

Oral-exam answer**Brain constraints**

For conventional brain plans, I focus on brainstem, optic nerves, optic chiasm, cochlea, and normal brain. For SRS, I care heavily about conformity, gradient, and normal brain V_{12} .

Oral-exam answer**Prostate constraints**

For prostate, I focus on rectum and bladder high-dose volumes, bowel if pelvic nodes are treated, femoral heads, and penile bulb when relevant.

Oral-exam answer**Re-irradiation**

There is no universal safe number. I review prior dose, overlap, fractionation, time interval, cumulative BED/EQD2, and serial OAR tolerance.

Oral-exam answer**Palliation**

Palliation shifts the balance toward simplicity and symptom relief, but serial OARs still matter. Common regimens include 8 Gy $\times 1$, 20 Gy in 5, and 30 Gy in 10.

Oral-exam answer**EUD**

EUD is the uniform dose that would produce the same biological effect as the nonuniform dose distribution. The a parameter determines whether the metric emphasizes mean dose, hot spots, or cold spots.

Oral-exam answer**CSI setup**

Classic CSI uses opposed lateral cranial fields matched to one or more posterior spine fields. The key issue is junction management, usually with feathering.

Oral-exam answer**Mantle fields**

Mantle fields are historical large AP/PA supradiaphragmatic fields for Hodgkin lymphoma, covering cervical, supraclavicular, mediastinal, hilar, and axillary regions with custom shielding.

Oral-exam answer**Isocenter versus calculation point**

Isocenter is a geometric delivery reference. The calculation point is a dosimetric point. In IMRT/VMAT, prescription and evaluation are usually volume-based, not point-based.

Oral-exam answer**Dose grid**

Use the clinically validated dose grid. Routine IMRT/VMAT may use 2–3 mm, but small targets and steep gradients require finer grids.

Oral-exam answer**A plan passes gamma. Is it good?**

Not necessarily. Gamma is a QA screen. I still need target coverage, OAR DVHs, secondary dose verification, deliverability, robustness, and clinical approval.

12 Final High-Yield Checklist

High-yield point

If you can say the following things clearly, you are in strong shape for Dosimetry 2 oral exam:

1. I can choose between 3DCRT, static IMRT, and VMAT based on anatomy, OARs, robustness, and intent.
2. I can explain forward versus inverse planning.
3. I know what makes VMAT delivery and QA unique.
4. I know TG-218-style VMAT QA criteria.
5. I can distinguish serial and parallel organs.
6. I know key dose-limit anchors for prostate, abdomen, head and neck, thorax, and brain.
7. I can explain how dose limits change in re-irradiation and palliation.
8. I can explain EUD and the a parameter.
9. I can describe CSI setup and junction feathering.
10. I can describe historical mantle fields.
11. I know isocenter versus calculation point.
12. I know dose grid principles.
13. I can troubleshoot common planning failures.

Common trap

The single best oral-exam closing line

The best plan is not the most complex plan. It is the simplest robust plan that safely achieves the clinical goal.

13 References

References

- [1] Miften M, Olch A, Mihailidis D, et al. Tolerance limits and methodologies for IMRT measurement-based verification QA: recommendations of AAPM Task Group No. 218. *Medical Physics*. 2018;45(4):e53–e83.
- [2] Zhu TC, Stathakis S, Clark JR, et al. Report of AAPM Task Group 219 on independent calculation-based dose/MU verification for IMRT. *Medical Physics*. 2021;48(10):e808–e829.
- [3] Ezzell GA, Galvin JM, Low D, et al. IMRT commissioning: multiple institution planning and dosimetry comparisons, a report from AAPM Task Group 119. *Medical Physics*. 2009;36:5359–5373.
- [4] Benedict SH, Yenice KM, Followill D, et al. Stereotactic body radiation therapy: the report of AAPM Task Group 101. *Medical Physics*. 2010;37(8):4078–4101.
- [5] Klein EE, Hanley J, Bayouth J, et al. Task Group 142 report: quality assurance of medical accelerators. *Medical Physics*. 2009;36(9):4197–4212.
- [6] Kry SF, Feygelman V, Balter P, et al. AAPM Task Group 329: reference dose specification for dose calculations: dose-to-water or dose-to-muscle? *Medical Physics*. 2020;47(3):e52–e64.
- [7] Bentzen SM, Constine LS, Deasy JO, et al. Quantitative analyses of normal tissue effects in the clinic: QUANTEC. *Int J Radiat Oncol Biol Phys*. 2010;76(3 Suppl):S3–S9.
- [8] Marks LB, Bentzen SM, Deasy JO, et al. Radiation dose-volume effects in the lung. *Int J Radiat Oncol Biol Phys*. 2010;76(3 Suppl):S70–S76.
- [9] Kirkpatrick JP, van der Kogel AJ, Schultheiss TE. Radiation dose-volume effects in the spinal cord. *Int J Radiat Oncol Biol Phys*. 2010;76(3 Suppl):S42–S49.
- [10] Deasy JO, Moiseenko V, Marks L, Chao KSC, Nam J, Eisbruch A. Radiotherapy dose-volume effects on salivary gland function. *Int J Radiat Oncol Biol Phys*. 2010;76(3 Suppl):S58–S63.
- [11] Kavanagh BD, Pan CC, Dawson LA, et al. Radiation dose-volume effects in the stomach and small bowel. *Int J Radiat Oncol Biol Phys*. 2010;76(3 Suppl):S101–S107.
- [12] Niemierko A. Reporting and analyzing dose distributions: a concept of equivalent uniform dose. *Medical Physics*. 1997;24(1):103–110.
- [13] Otto K. Volumetric modulated arc therapy: IMRT in a single gantry arc. *Medical Physics*. 2008;35(1):310–317.
- [14] Ling CC, Zhang P, Archambault Y, et al. Commissioning and quality assurance of RapidArc radiotherapy delivery system. *Int J Radiat Oncol Biol Phys*. 2008;72(2):575–581.
- [15] International Commission on Radiation Units and Measurements. Prescribing, Recording, and Reporting Photon-Beam Intensity-Modulated Radiation Therapy. ICRU Report 83. 2010.